



Document: Phase 1 Domain Review & Synthetic Railway Data Specification

Workstream Owner: Darshini (Domain Modeling & Synthetic Data)

Branch: feature/domain-data

Event: Smart India Hackathon 2026

Integration Owner: Aryan

Status: Complete & Verified (Ready for Domain Freeze)

1. Executive Summary & Purpose

This document formalizes the Indian Railways operational domain modeling, shared data schemas, and reproducible synthetic corridor scenarios for **Phase 1** of Project Pashupatastra. All downstream modules (CP-SAT Optimizer by Tyagi, ML Scorer by Ayush, Disruption Simulation by Tirth, Dashboard UI by Archit, and FastAPI Backend by Aryan) consume the models and fixtures documented herein.

2. Domain Review & Decisions (Domain Freeze Checkpoint)

2.1 Track Modeling

Decision: Retain flat string `track_id` (e.g. "UP-1" , "DOWN-1") as the primary key in CP-SAT constraints for solver performance, while attaching structured corridor metadata (section name, mileage, direction, speed limit) in the data layer for UI filtering.

2.2 Maintenance Work Types

Standardized 6 concrete Indian Railways maintenance categories mapped to physical operations and machinery:

Work Category	Operational Description	Typical Duration	Required Machinery / Gang	Priority Range
TRACK_RENEWAL	Complete Track Renewal (CTR), Sleeper & Rail renewal	180 – 240 min	BCM_MACHINE_01 (Ballast Cleaner)	0.85 – 0.95
BALLAST_TAMPING	Track packing, alignment & geometry correction	90 – 150 min	CSM_TAMPER_01 (Continuous Tamper)	0.70 – 0.85
OHE_MAINTENANCE	Overhead Electrification inspection & isolations	75 – 120 min	TOWER_WAGON_01 (Inspection Unit)	0.65 – 0.80
SIGNALLING_INTERLOCKING	Point machine overhaul, track circuits, axle counters	60 – 120 min	S_T_GANG_01 (Signal & Telecom Gang)	0.60 – 0.75
ROUTINE_INSPECTION	USFD ultrasonic rail flaw detection & foot patrol	45 – 90 min	Manual Trolley Gang	0.35 – 0.50
EMERGENCY_REPAIR	Rail fracture, weld failure, snapped OHE wire	90 – 150 min	Emergency Flying Squad	0.95 – 0.99

2.3 Resource & Equipment Constraints

Decision: Avoid multi-resource bin-packing that bloats solver complexity. Shared heavy machines (BCM, CSM Tamper, Tower Wagon) are constrained using `mutual_exclusion_group` intervals, preventing simultaneous overlapping use across different tracks.

2.4 Train Timetable & Operational Possession Windows

Train movements are modeled via discrete available **Possession Windows** per track over a 24-hour horizon (1440 min):

Window ID	Track	Window Type	Clock Time	Minute Interval	Operational Purpose
POS-UP1-NIGHT	UP-1	NIGHT_TRAFFIC_BLOCK	00:30 – 05:00	0030 – 0300 (270 min)	Primary heavy renewals & tamping
POS-UP1-MIDDAY	UP-1	MIDDAY_MAINTENANCE	11:30 – 14:30	0690 – 0870 (180 min)	Inspections & OHE maintenance
POS-UP1-EVENING	UP-1	EVENING_OFF_PEAK	21:00 – 23:30	1260 – 1410 (150 min)	Signalling & minor adjustments
POS-DN1-NIGHT	DOWN-1	NIGHT_TRAFFIC_BLOCK	00:30 – 05:00	0030 – 0300 (270 min)	Primary heavy renewals & tamping
POS-DN1-MIDDAY	DOWN-1	MIDDAY_MAINTENANCE	11:30 – 14:30	0690 – 0870 (180 min)	Inspections & OHE maintenance
POS-DN1-EVENING	DOWN-1	EVENING_OFF_PEAK	21:00 – 23:30	1260 – 1410 (150 min)	Signalling & minor adjustments

3. Synthetic Scenario Datasets

3.1 Scenario A: Baseline Mainline Corridor (`corridor_a_blocks.json`)

Corridor: `CORRIDOR_A` (UP-1, DOWN-1) | **Candidates:** 12 Blocks | **Horizon:** 1440 min | **Headway:** 15 min

Block ID	Track	Asset ID	Work Category	Dur	Pri	Risk	Allowed Window	Machine Group	Dependency
BLK-001	DOWN-1	AST-DN-OHE-013	OHE_MAINTENANCE	120 m	0.79	0.63	11:00 – 15:30 (0660-0930)	TOWER_WAGON_01	—
BLK-002	DOWN-1	AST-DN-RAI-015	BALLAST_TAMPING	105 m	0.50	0.50	00:30 – 06:00 (0030-0360)	CSM_TAMPER_01	—
BLK-003	DOWN-1	AST-DN-RAI-014	TRACK_RENEWAL	210 m	0.59	0.66	00:30 – 06:00 (0030-0360)	BCM_MACHINE_01	—
BLK-004	DOWN-1	AST-DN-RAI-009	BALLAST_TAMPING	150 m	0.59	0.59	04:00 – 07:30 (0240-0450)	CSM_TAMPER_01	BLK-003 
BLK-005	UP-1	AST-UP-OHE-004	OHE_MAINTENANCE	75 m	0.56	0.52	00:30 – 23:30 (0030-1410)	TOWER_WAGON_01	—
BLK-006	DOWN-1	AST-DN-OHE-016	OHE_MAINTENANCE	120 m	0.54	0.49	00:30 – 23:30 (0030-1410)	TOWER_WAGON_01	—
BLK-007	DOWN-1	AST-DN-OHE-011	OHE_MAINTENANCE	120 m	0.79	0.53	11:00 – 15:30 (0660-0930)	TOWER_WAGON_01	—
BLK-008	UP-1	AST-UP-RAI-001	TRACK_RENEWAL	210 m	0.78	0.77	00:30 – 06:00 (0030-0360)	BCM_MACHINE_01	—
BLK-009	UP-1	AST-UP-TUR-007	SIGNALLING_INTERLOCK	105 m	0.42	0.44	00:30 – 23:30 (0030-1410)	S_T_GANG_01	—
BLK-010	UP-1	AST-UP-RAI-008	ROUTINE_INSPECTION	45 m	0.42	0.46	11:00 – 15:30 (0660-0930)	—	—
BLK-011	UP-1	AST-UP-SIG-002	SIGNALLING_INTERLOCK	75 m	0.65	0.59	00:30 – 23:30 (0030-1410)	S_T_GANG_01	—

Block ID	Track	Asset ID	Work Category	Dur	Pri	Risk	Allowed Window	Machine Group	Dependency
BLK-012	UP-1	AST-UP-TUR-006	SIGNALLING_INTERLOCK	75 m	0.57	0.50	00:30 – 23:30 (0030-1410)	S_T_GANG_01	—

3.2 Scenario B: High Density 4-Track Trunk Corridor (`corridor_b_dense.json`)

Corridor: `CORRIDOR_B_DENSE` | **Tracks:** `UP-1`, `UP-2`, `DOWN-1`, `DOWN-2` | **Candidates:** 24 Blocks

Simulates heavy quadruple trunk routes (e.g. New Delhi – Kanpur). Contains 4 competing track machine groups with tight possession windows to test CP-SAT optimization and solver trade-offs under saturation.

3.3 Scenario C: Disruption & Recovery Testbed (`corridor_c_disrupted.json`)

Corridor: `CORRIDOR_C_DISRUPTED` | **Tracks:** `UP-1`, `DOWN-1` | **Candidates:** 14 Blocks

Pre-configured with **2 COMMITTED blocks** (locked on `DOWN-1` from 01:00 to 03:15). Ready for Tirth's disruption simulator to inject `TRACK_UNAVAILABLE` on `UP-1` from 02:00 to 05:00 and test CP-SAT recovery.

4. Code & Artifact Checklist

- ✓ **DOMAIN_REVIEW.md:** `C:\Users\Hp\.gemini\antigravity\scratch\pashupatastra\DOMAIN_REVIEW.md`
- ✓ **Data Generator:** `C:\Users\Hp\.gemini\antigravity\scratch\pashupatastra\backend\app\data\generator.py`
- ✓ **Domain Models:** `C:\Users\Hp\.gemini\antigravity\scratch\pashupatastra\backend\app\data\models.py`
- ✓ **JSON Fixtures:** `backend/app/data/fixtures/` (`corridor_a_blocks.json` , `corridor_b_dense.json` , `corridor_c_disrupted.json`)
- ✓ **Unit Test Suite:** `backend/tests/test_generator.py` (**10/10 tests passed** in 0.049s)
- ✓ **CLI Runner:** `generate_data.py` (Run `python generate_data.py --scenario A`)

Repository Sync: All files are mirrored at `D:\priyadarshini projects\pashupatastra\` and `C:\Users\Hp\.gemini\antigravity\scratch\pashupatastra\` .